Biosensors and Bioelectronics 182 (2021) 113190

Contents lists available at ScienceDirect

Biosensors and Bioelectronics

journal homepage: http://www.elsevier.com/locate/bios

DEPIS: A combined dielectrophoresis and impedance spectroscopy platform for rapid cell viability and antimicrobial susceptibility analysis

Pragya Swami, Ayush Sharma, Satyam Anand, Shalini Gupta *

Dept. of Chemical Engineering, Indian Institute of Technology, Delhi, 110016, India

ARTICLE INFO

Keywords:
Impedance
Dielectrophoresis
Antimicrobial susceptibility
Antimicrobial resistance
Bacterial diagnosis

ABSTRACT

Antimicrobial resistance (AMR) is caused by inappropriate or excessive antibiotic consumption. Early diagnosis of bacterial infections can greatly curb empirical treatment and thus AMR. Current diagnostic procedures are time-consuming as they rely on gene amplification and cell culture techniques that are inherently limited by the doubling rate of the involved species. Further, biochemical methods for species identification and antibiotic susceptibility testing for drug/dose effectiveness take several days and are non-scalable. We report a real-time, label-free approach called DEPIS that combines dielectrophoresis (DEP) for bacterial enrichment and impedance spectroscopy (IS) for cell viability analysis under 60 min. Target bacteria are captured on interdigitated electrodes using DEP (30 min) and their antibiotic-induced stress response is measured using IS (another 30 min). This principle is used to generate minimum bactericidal concentration (MBC) plots by measuring impedance change due to ionic release by dying bacteria in a low conductivity buffer. The results are rapid since they rely on cell death rather than cell growth which is an intrinsically slower process. The results are also highly specific and work across all bactericidal antibiotics studied, irrespective of their cellular target or drug action mechanism. More importantly, preliminary results with clinical isolates show that methicillin-susceptible *Staphylococcus aureus* (MSSA) can easily be differentiated from methicillin-resistant *S. aureus* (MRSA) under 1 h. This rapid cell analyses approach can aid in faster diagnosis of bacterial infections and benefit the clinical decision-making process for antibiotic treatment, addressing the critical issue of AMR.

1. Introduction

Bacterial infections and their associated challenges pose a major threat to human life as they cause over 8 million deaths worldwide per annum (Ten et al., 2017; Christiansen, 2018; Superbug Solutions Platform, 2019). These mortalities can be greatly reduced by faster drug screening and rapid susceptibility profiling of bacterial targets. Present clinical techniques for antibiotic susceptibility testing (AST) such as Kirby-bauer disk diffusion, E-test, and broth dilution assays depend on bacterial growth inhibition in the presence of antibiotics (Syal et al., 2017; Jorgensen and Ferraro, 2009). Although these methods are simple and accurate, they are tedious to perform and have a high turnaround time of 24 h or more. This delay promotes empirical antibiotic therapy and often derails proper medical treatment causing rampant rise in AMR in several critical bacterial strains (Ventola, 2015; Llor and Bjerrum, 2014). Recent estimates predict annual AMR-related deaths to exceed 10 million by 2050 which is higher than cancer, diabetes and cardiovascular diseases (O'Neill, 2014). Thus, there is an urgent unmet need to develop new strategies that allow rapid diagnosis of bacterial infections and tuning of therapeutic dosages at the earliest possible stage of treatment.

With the increasing clinical demand of rapid diagnosis, there are various USFDA approved phenotypic AST techniques already available in the market (Dietvorst et al., 2020). These include MALDI-TOF mass spectrometry (Hou et al., 2019), turbidity and fluorescence-based Vitek and MicroScanWalkAway analyzers (Felmingham and Brown, 2001), and multiplexed automated digital microscopy systems (Chantell, 2015). These techniques are useful and accurate but are intrinsically limited by the doubling rate of the pathogens. They also require trained manpower to operate the complex equipment. Nucleic acid amplification-based polymerase chain reaction (PCR) is another extremely sensitive genotypic method for direct detection of resistant bacterial isolates (Fluit et al., 2001). This too, however, requires prior knowledge of genetic information, expensive reagents, complex amplification steps, and skilled personnel for operations and maintenance. More recently, new phenotypic AST methods have emerged such as

* Corresponding author.
E-mail address: shalinig@chemical.iitd.ac.in (S. Gupta).

https://doi.org/10.1016/j.bios.2021.113190
Received 3 November 2020; Received in revised form 11 March 2021; Accepted 18 March 2021
Available online 26 March 2021

bacterial growth monitoring using microscopy (Choi et al., 2013; Fredborg et al., 2013; Li et al., 2014), biomass and density monitoring using microcantilevers (Longo et al., 2013) and microchannel resonance (Godin et al., 2007), rotation and motion detection using magnetic (Sinn et al., 2012) and electric fields (Rohani et al., 2018) (EFs), and viability testing using fluorescent protein dyes (Webb et al., 2001; Johnson and Criss, 2013) or 16s rRNA count (Mohan et al., 2011). These methods take anywhere from 3 to 8 h for AST and require extensive optimization, sample preparation and pre-treatment.

To overcome these present limitations and challenges, scientists have focused on bacterial analyses using impedance-based electrical sensing (Furst and Francis, 2019; Xu et al., 2016; Yang and Bashir, 2008; Levo-Bueno et al., 2020; Spencer et al., 2020). Impedance spectroscopy (IS) is a simple, rapid, robust, label-free and high-throughput method that allows probing cellular response in real-time and can be miniaturized, automated and multiplexed for viable sensing of pathogens in remote settings (Lei, 2014; Gnaim et al., 2020). The current bacterial impedance sensors are of two kinds. One, that detect the presence of bacteria directly in suspensions (Yang, 2008) or after they have been captured on a surface using bioaffinity receptors such as antibodies (Dastider et al., 2013, 2015; Suehiro et al., 2006; Mallén-Alberdi et al., 2016). Second are those that differentiate between live and dead bacteria by measuring cell growth/death either directly or through the release of certain metabolites in response to an external stress (Liu et al., 2018; Li et al., 2018; Gomez-Sjoberg, 2005). While the first kind of sensors require complex chip designs and electrode surface modifications to enhance impedance signal output, the second category of sensors suffer from poor sensitivity and delayed response because of dilutional and diffusional limitations in measuring bacteria-related events (*e.g.* drug-induced release of metabolites) that take place in the bulk far away from the detection zone. There is also lack of experimental evidence on low conductivity buffers to study bacterial cytoplasmic impedance response within a practical frequency limit (<10 MHz) (Yang, 2008; Couniot et al., 2014).

A few recent studies have reported impedance-based cell viability assays that yield a yes/no type result in the presence of antibiotics (Mannoor et al., 2010; Safavieh et al., 2017; Lillehoj et al., 2014). While these studies can identify correct antibiotics for pathogens through cell-drug interaction measurements, they do not yield a bactericidal/bacteriostatic drug concentration-dependent dose response curve, commonly known as MIC (minimum inhibitory concentration) or MBC (minimum bactericidal concentration) plot, when compared to conventional antibiograms (Liu et al., 2004). We believe this may be so because these studies use high conductivity media, typically Luria Bertani broth or phosphate buffered saline (PBS) buffer, that give rise to a large background noise. Also, the location of bacteria on the chip may not be as close to the detection zone as desired from an EF sensing point-of-view. Our group recently demonstrated that reversible patterning of bacteria onto high-throughput microarrays using dielectrophoresis (DEP) can be used for rapid cellular phenotyping when combined with IS or optical microscopy (Goel et al., 2018). The sensitivity of detection was enhanced through the novel use of a zwitterionic buffer HEPES (Pal et al., 2016; Khandelwal et al., 2019).

In this paper, we further extended our approach to carry out label-free cell viability testing, antibiotic mechanism-based drug screening, MBC_{50} determination, and resistant versus susceptible bacterial profiling in real-time under 1 h. This was achieved by combining DEP with IS (dubbed DEPIS) to accelerate cell viability analysis in the presence of different bactericidal antibiotics. The drugs were chosen to cover a wide spectrum of cellular targets (*e.g.* cell membrane, cell wall *etc.*) and antibiotic action mechanisms to demonstrate a generalized approach. Similarly, bacterial species included both Gram-negative and Gram-positive susceptible and resistant strains. To keep the setup basic, studies were performed on standard reusable interdigitated electrodes (IDEs) with large gap sizes (50 μm) without any fluid flow. Our entire hypothesis was pivoted on the fact that intracellular ionic release by living cells under antibiotic stress correlates with their viability and therefore should, in principle, yield much faster results than conventional systems that monitor cell growth (not cell death) which is an inherently slower process. Although this combined approach (DEP and IS) has been used by the various other researchers to measure the impedance of whole cells after their collection on electrodes (Liu et al., 2018; Dastider et al., 2018; Sabounchi et al., 2008; Moral-zamora et al., 2015), ours is the first study in which DEP-enhanced IS used to measure antibiotic-induced ionic release for cell viability and antimicrobial susceptibility testing.

Theory. ***(i) Dielectrophoresis.*** DEP is a phenomenon in which a force (given by eq. (1)) is exerted on a particle suspended in a medium under the influence of a non-uniform EF (Pethig, 2010).

$$F_{DEP} = 2\pi\varepsilon_m r^3 Re(f_{cm})\left(\nabla E^2\right) \tag{1}$$

The DEP force depends on the particle radius (*r*), absolute permittivity of the medium (ε_m), gradient of the square of the EF (∇E^2) and the real part of the Clausius–Mossotti (CM) factor written as (eq. (2)),

$$f_{cm} = \frac{\varepsilon_p^* - \varepsilon_m^*}{\varepsilon_p^* + 2\varepsilon_m^*} \tag{2}$$

Here, ε^* is the complex permittivity defined as $\varepsilon^* = \varepsilon\varepsilon_0 - \frac{\sigma j}{\omega}$, where, σ is conductivity, $j = \sqrt{-1}$ is imaginary factor, ε_0 is the vacuum permittivity (8.85×10^{-12} F/m), ε is relative permittivity and $\omega = 2\pi f$ is the angular frequency of the applied EF. The subscripts *p* and *m* stand for particle and medium, respectively. The non-uniformity in the EF prompts the particles to go to areas of maximum or minimum EF intensity depending on the sign of the CM factor. For $Re(f_{cm}) > 0$, the particles being more polarizable than the medium experience positive DEP (pDEP) and get pulled towards areas of high EF intensity, and vice versa.

(ii) Impedance. This is a frequency-dependent phenomenon where a small voltage (*V*) is applied across a system and its electrical current (*I*) is measured. The impedance (*Z*) is then calculated as the ratio of the applied voltage to the measured current and is a function of resistance (*R*), capacitance (*C*) and the applied frequency (eq. (3)).

$$Z = \frac{V}{I} = R + \frac{1}{j\,\omega\,C} \tag{3}$$

The impendence can also be reported in terms of complex permittivity of the system and its geometrical cell constant, *K* (m^{-1}) (eq. (4)),

$$Z = \frac{K}{j\,\omega\,\varepsilon^*} \tag{4}$$

In case of bacterial cells suspended in a medium, each cell can be modeled as a dielectric particle that gets polarized in the presence of the applied EF. These cells then contribute additive dipole moments in the overall impedance measurement of the system (Valero et al., 2010). The overall impedance of the bacterial suspension can be written as (eq. (5)),

$$Z = \frac{K}{j\,\omega\,\varepsilon_s^*} \tag{5}$$

where, ε_s^* is the equivalent complex permittivity of the suspension containing the medium and the cells that occupy a certain volume fraction (ϕ) in the medium. This equivalent complex permittivity of the cell suspension can be calculated by Maxwell's mixture theory (Sun et al., 2008) as (eq. (6)),

$$\varepsilon_s^* = \varepsilon_m^*\left[\frac{1 + 2\phi f_{cm}}{1 - \phi f_{cm}}\right] \tag{6}$$

From eqs. (1), (5) and (6), it can be concluded that DEP and impedance are both induced dipole phenomena, ruled by the CM factor. In case of impedance, the induced dipoles are detected without perturbing the system through use of a low primary voltage (Barbero et al., 2005). In DEP, the induced dipoles in the cells experience a force leading to the displacement of cells towards high or low EF gradient.

2. Experimental section

2.1. Materials

Viable bacterial strains of *Salmonella typhi* (*S. typhi*) and *Staphylococcus aureus (S. aureus)* in glycerol stock stored at −20 °C (Kusuma School of Biological Sciences and Dept. of Biochemical Engineering and Biotechnology, IIT Delhi); Clinical isolates of methicillin-susceptible and methicillin-resistant *S. aureus* in glycerol stock stored at −20 °C (Global hospitals, Hyderabad, India); Clinical isolates of levofloxacin-susceptible and levofloxacin-resistant *Enterococcus faecalis* (*E. faecalis)* in blood agar plate stored at 4 °C (Maulana Azad Medical College, New Delhi, India); Sterile nylon filter (13 mm/0.2 μm), absolute ethanol (99% purity), sodium hydroxide, sodium chloride and PBS tablets (Merck, India); Sushi S3 peptide (HAEHKVKIGVEQKYGQFPQGTEV-TYTCSGNYFLMC, M.W. 4083 Da, purity >95%) (Genemed Synthesis, San Antonio, USA); Polydimethylsiloxane (PDMS) elastomer kit (Dow Corning, USA); Polymyxin B sulfate (PMB), bacitracin, methicillin sodium, LIVE/DEAD BacLight bacterial viability kit (L7012) and 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES) (Sigma Aldrich, India); Levofloxacin (SRL, India); Luria Bertani (LB) agar miller, LB broth and sterile Petri dishes (Himedia Labs, India); Sterile spreader (Tarson, India); Isopropyl alcohol (IPA) and acetone (Fisher Scientific, India); Platinum-coated interdigitated electrodes (IDEs) on glass with 40 electrode fingers (20 pairs) of 3.5 mm length, 50 μm width and 50 μm spacing (Micrux Technologies, Spain). Ultrapure deionized Milli-Q water of ~18.2 MΩ cm resistivity (Millipore, India) was used to prepare all the buffers. All experiments were performed at room temperature (~25 °C) and bacterial sample preparations and plating experiments were always carried out under sterile laminar air flow conditions after autoclaving the reagents.

2.2. Methods

2.2.1. Buffer preparation

15 mM HEPES buffer was prepared by reconstituting 1.43 g of HEPES powder in 400 mL MilliQ water and the pH was adjusted to pH 7.4 by adding 1.9 mL of 1 M NaOH. 5 M NaCl stock was prepared by reconstituting 1.16 g of NaCl powder in 10 mL of MilliQ water and serially diluted (with MilliQ water) to obtain different molarity solutions in the 0.025–125 mM NaCl concentration range. These different molarity solutions were then mixed with HEPES or any other buffer in 1:39 v/v ratio to obtain samples for the conductivity study. All the buffers were filtered through sterile nylon filters to remove any contamination. Buffer conductivities were measured using the Mettler Toledo Lab-731 ISM conductivity meter and pH values were obtained using the Eutech Instruments pH-7200 m.

2.2.2. Bacterial suspensions

100 μL of 50% bacterial-glycerol suspension or alternately, one colony from the agar plate, was transferred into 3 mL LB broth taken in 15 mL falcon tube and grown in a B.O.D cum shaker incubator (Ultra Lab instruments, New Delhi) for 12–15 h at 37 °C and 130 rpm. A 1 mL aliquot from this stock of grown bacterial suspension was taken in a microcentrifuge tube, centrifuged twice at 10,375 rcf for 10 min and the supernatant was replaced with the buffer. The optical density (O.D.) of the suspension was measured at 600 nm wavelength using a UV–visible spectrophotometer (Shimadzu UV-2600) and the stock was serially diluted with the buffer to the desired final concentration of 10^7 CFU/mL bacteria corresponding to an optical density (O.D.) of 0.02 for a 1 mm path length.

2.2.3. Antimicrobial activity assays

Cell viability tests against different antibiotics were performed by plating, O.D. and impedance methods. For this, antibiotic stock suspensions of different molar concentrations were made in 15 mM HEPES at pH 7.4 and mixed with the bacterial suspension (10^7 CFU/mL) in 1:39 v/v ratio. The unreacted drug was removed after 30 or 60 min time intervals by centrifuging the sample at 10,375 rcf for 10 min. The supernatant was replaced by 1 mL of 15 mM HEPES buffer and the suspension was then serially diluted with 15 mM HEPES to make a 10^5 CFU/mL cell suspension. A 10 μL aliquot of this suspension was spread on a sterile LB agar plate with the help of a sterile spreader. To prepare the agar plates, a 40 g/L LB agar miller stock solution was prepared in MilliQ water and ~12 mL of this was poured into each sterile Petri dish. The agar was then allowed to solidify at room temperature and the Petri dishes were placed in a B.O.D. cum shaker incubator (Ultra Lab instruments, New Delhi) for 16–20 h to free them of any contamination. The plates containing the bacterial spreads were immediately placed in a B.O.D. cum shaker incubator for 12–18 h and the bacterial colonies were counted the next day. The bacterial growth without any drug was taken as positive control and used for normalization of bacterial viability under all other conditions. Time-dependent O.D. measurements were separately performed at 600 nm wavelength over 60 min at 30 min time intervals using a UV–Visible spectrophotometer (Shimadzu UV-2600).

2.2.4. DEPIS platform

The experimental setup comprised of an IDE chip connected to both an impedance analyser (E4990A, Keysight) and a waveform function generator (33500B Agilent). Prior to use, the IDEs were cleaned with IPA, DI water and 70% v/v ethanol to remove any impurities and dried with nitrogen. Next, a PDMS slab of 5 mm (l) x 5 mm (w) x 1.6 mm (h) was prepared using a standard protocol (McDonald et al., 2000) and placed on top of the IDEs after punching a 4 mm diameter hole in its center. A 20 μL cell suspension with or without antibiotic was freshly prepared and immediately added to the PDMS chamber. The chamber was sealed by vacuum grease with the help of a glass cover slip and the first impedance reading was taken immediately after sample injection (time t = 0). The impedance analyser was then switched off and the function generator was used to apply sinusoidal AC signals across the suspension at 20 V_{pp} and 5 MHz frequency for pDEP bacterial capture. After 30 min, the function generator was switched off and the impedance analyser was again turned on to record readings over the next 30 min at 10 min time intervals. The final results were reported as differential impedance response (ΔI) with respect to the initial impedance value at t = 0 and plotted as a function of frequency or time. All impedance measurements were carried out at ± 100 mV in 1 kHz–10 MHz frequency range. The IDEs were rinsed as discussed above and reused until they wore off. All experiments were performed thrice and the results were reported as the mean ± 1 SD. Unpaired student's t-tests were performed between independent data sets using GraphPad Prism 8 software to check their statistical significance using the null hypothesis (*p*). The confidence interval for rejecting the null hypothesis was chosen as 95% ($p \leq 0.05$).

2.2.5. Fluorescence cell imaging

The bacterial cells were stained using the LIVE/DEAD viability kit containing Syto 9 and PI fluorescent dyes. For this, an aliquot of 1.5 μL of each dye was taken from its stock solution and added directly to 1 mL of bacterial suspension containing 10^7 cells/mL. After 20 min of incubation, antibiotic stocks were added to the dye-stained cell suspension and 20 μL of this mixture was further added to the IDE setup for 30 min of pDEP capture. The cell capturing process was monitored in real-time at 50× magnification using an Olympus BX-53 optical microscope fitted with an Olympus E3 CCD colour camera in either the brightfield or manual fluorescence mode (exposure target 235, exposure time 1000 ms, gain 1) using the FITC (for live green cells) and TXRED (for dead red cells) filters. The cell counting and background error clearance were done manually using the Image J software (version 1.48).

3. Results and discussion

3.1. Parameter selection and system optimization

Alternating EFs were applied in two stages across bacterial suspensions containing an antibiotic (Fig. 1). In the first step, live bacteria were dielectrophoretically captured from the bulk suspension at the electrode edges (where the EF intensity was highest) using a specified voltage and frequency for 30 min. Next, the antibiotic-induced electrophysiological stress response of the cells was measured by IS over a wide frequency range in a time-dependent manner from 30 to 60 min. Impedance readings were obtained only after lowering the voltage by 200x (DEP requires much higher voltage than IS), however, no loss in the number of cells was observed as a result of this lowering of EF. The number of cells in fact continued to rise, albeit at a relatively slower rate, due to gravity (Peclet number, $Pe = \frac{\frac{4}{3}\pi a^4 \Delta\rho g}{kT}$ ~1.5, where, $a = 1.1$ µm is the cell radius, $\Delta\rho = 0.1$ g cm^{-3} is the difference in density between a cell and the suspending medium, $k = 1.38 \times 10^{-23}$ m^2 kg s^{-2} K^{-1} is the Boltzmann constant and $T = 298$ K is temperature) while the cells that were already collected displayed Brownian motion as they diffused out from the electrode edges on the bottom plane, suggesting that they were not stuck to the surface. It may be noted that the rise in cell concentration was *not* due to cell division as we used a no growth buffer (mentioned ahead) and the bacteria remained in the stationary phase throughout the 1 h study (Fig. S1).

DEP enrichment was a crucial step to reduce the time of ionic transport to the detection zone (same as the capture electrodes in our case) as the antibiotic-induced ionic release was a continuous process that took place over the entire duration of 60 min and the ionic diffusion time through the bulk was estimated to be rather high (at least 10 min; see Table S1). The EF lines in our setup also only extended up to 1/16th of the total sample height (~100 µm). The pDEP frequency was chosen such that it allowed maximum live cell capture in a stipulated time and its value was determined using the well-known bacterial double shell model (see **SI** for details). The results obtained by plotting the effective bacterial CM factor as a function of frequency at various medium conductivities revealed that a medium conductivity of less than 0.4 S/m was essential to attain the pDEP condition (Fig. S2). Further, a pDEP capture frequency of 5 MHz was fixed for all our subsequent experiments as the maximum bacterial capture at this condition took place between the 5–10 MHz frequency range. Among the various low conductivity buffers tested (<0.4 S/m), 15 mM HEPES at pH 7.4 (0.04 S/m) was found to give the best results. This buffer not only showed a high ΔI response as a function of salt concentration (Fig. 2a) but also caused the least amount of stress to the bacterial cells in the absence of drugs (Fig. 2b). In general, low conductivity buffers can put undue stress on the cells making them leaky due to a low osmolarity condition, however, HEPES is a Good's buffer that is known to have a unique electrostatic crosslinking property with the cell wall that allows cells to bear high turgor pressures (Rani and Patri, 2019; Suo et al., 2007). The limit of detection (LOD) and the sensitivity of the sensor using 15 mM HEPES buffer was found to be 0.1 mM NaCl (corresponding to a ΔI of ~7 Ω) and 252 Ω/mM, respectively (Fig. S3). At our identified operating conditions, the CM factors for live and dead bacteria were calculated to be 0.63 and −0.24, respectively, using the double shell model (see **SI**), indicating that only live cells could be captured by pDEP at the electrode edges, not the dead ones. This was further confirmed experimentally as discussed ahead.

Next the optimal time for pDEP capture in 15 mM HEPES was studied

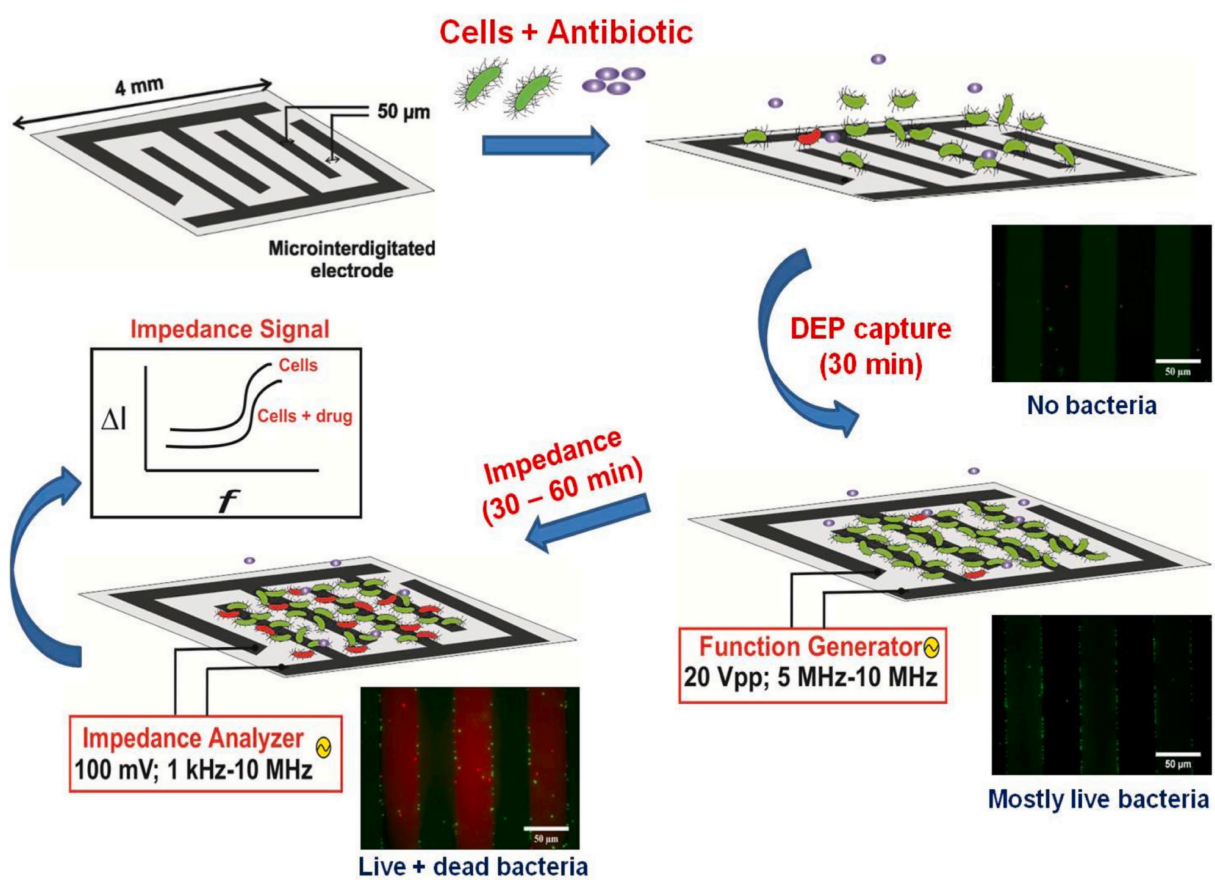

Fig. 1. Schematic of the DEPIS biosensor illustrating the chronological sequence of events involved in the detection process. The insets depict actual fluorescence micrographs of the bottom electrodes captured at various time points. Here, the live bacteria are green and the dead ones as red. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

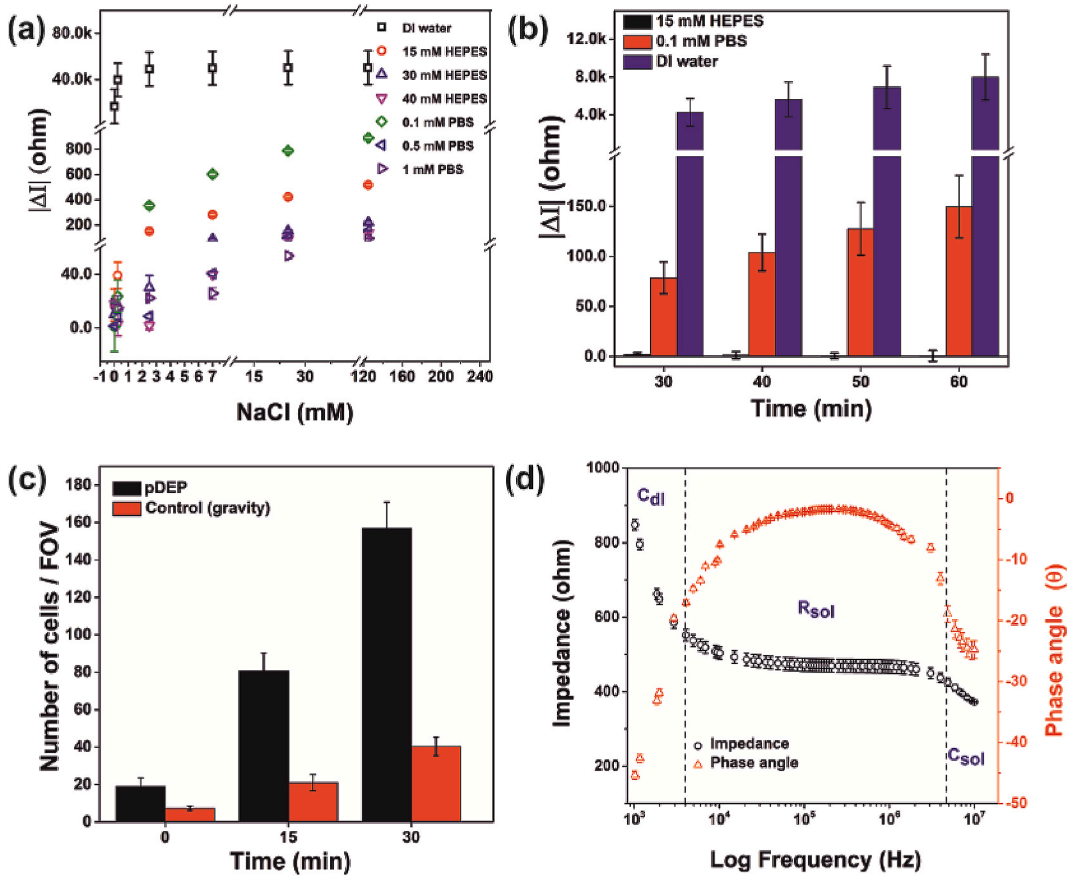

Fig. 2. **(a)** Differential impedance values ($|\Delta I|$) at 100 kHz frequency for several NaCl concentrations in low conductivity media: DI water (0.0001 S/m), 0.1 mM PBS (0.02 S/m), 15 mM HEPES (0.04 S/m), 30 mM HEPES (0.075 S/m), 40 mM HEPES (0.1 S/m), 0.5 mM PBS (0.1 S/m) and 1 mM PBS (0.18 S/m). **(b)** The $|\Delta I|$ response at 100 kHz measured as a function of time for *S. typhi* bacteria (10^7 cfu/mL) suspended in different media. The signal was indicative of the number of ions released by the bacteria. **(c)** Time-resolved pDEP capture (20 V_{pp}, 5 MHz) of *S. typhi* bacteria (10^7 cfu/mL) at the electrode surface. Here, one field of view (FOV) represents 37,000 μm^2 surface area. **(d)** Measured impedance and phase angle spectra of 15 mM HEPES buffer in the 1 kHz to 10 MHz frequency range. Lines *F1* and *F2* indicate analytically-derived dominating impedance factors in this frequency domain.

to reduce the overall time of our assay. It was found that the number of cells captured on the surface increased proportionally to the time of applied EF as expected. After 30 min, ~27% of the total bacteria present in the sample were concentrated near the electrode edges which translated into a surface density of 4240 ± 375 cells/mm^2 (see Fig. 2c & **SI** for more details). This was 4x higher than the surface density achieved by gravitational settling alone. Further, approx. 99% cells remained viable at the end of the 60 min process (Fig. S1) and no drift in signal due to Joule heating or bacterial growth was observed in this time (Fig. S4). The latter revealed that the bacteria remained in the stationary phase in our no growth buffer. Application of EF beyond 30 min led to further cell enrichment, however, the DEP time was restricted to 30 min in our experiments as it was found to give sufficient signal enhancement in the subsequent step. A shorter EF exposure time also helped minimize solvent evaporation and any leakage-related artefacts, that usually become significant in low volume setups (Gallo-Villanueva et al., 2014).

The optimum frequency for conducting the impedance experiments was estimated from a recently reported analytical model that takes into account the geometrical parameters of the electrodes and the electrical characteristics of the medium (Couniot et al., 2015). This model helps to calculate the cut-off frequency limits at which a system undergoes changes in its underlying polarization behaviour. For instance, the cut-off frequency limit F_1 below which the system impedance is dominated by the double layer capacitance at the electrode-solution interface (C_{dl}) can be estimated by eq. (7), while the upper frequency limit F_2 after which the capacitance of the solution (C_{sol}) controls the overall impedance of the system is given by eq. (8).

$$F_1 = \frac{1}{\pi} \cdot \frac{\lambda \sigma_m}{A\ \varepsilon \varepsilon_0} \cdot \frac{1}{K} \tag{7}$$

$$F_2 = \frac{1}{2\pi} \cdot \frac{\sigma_m}{\varepsilon \varepsilon_0} \tag{8}$$

Between these two frequency cut-offs, the impedance behaviour of the system is governed mainly by the solution resistance (R_{sol}). Here, λ indicates Debye length and A is the electrode area (half of the total area). For our system with $\lambda = 5.28$ nm, $\varepsilon = 80$, $A = 6.28\ \mu m^2$, $1/K = 0.058$ m and $\sigma_m = 0.04$ S/m, the values of F_1 and F_2 for 15 mM HEPES buffer were found to be 3.5 kHz and 9 MHz, respectively, which were in excellent agreement with our experimentally measured impedance and phase angle spectra (Fig. 2d). Since we were interested in measuring the ionic release by bacteria, all our impedance spectra were recorded within this frequency domain at 100 kHz where the impedance change mainly corresponded to the change in R_{sol} ($\theta \sim 2^\circ$) and not the change in C_{dl} due to ionic release.

3.1.1. *Cell viability testing and dose-response curves*

Once the operating parameters for our experiments were identified, we moved on to our main objective, i.e., the label-free viability testing of different bacteria against a wide range of antibiotics. For this purpose, several susceptible and resistant strains of both Gram-negative and Gram-positive bacteria were chosen and their interaction was studied with different classes of bactericidal drugs selected based on their cellular targets and mechanisms of action (list in Table 1). Polymyxin B

Table 1
List of bacterial strains and antibiotics used in our cell-drug interaction studies.

Bacteria			Bactericidal drug		Mechanism of drug action/resistance	Ref.
Type	Strain	Viability	Name	Cellular Target		
Gram-negative	*S. typhi*	Susceptible	Polymyxin B sulfate (PMB)	Cell membrane	Disrupts the outer cell membrane	Poirel et al. (2017)
		Susceptible	Sushi S3 peptide (sushi)			(Goel et al., 2018; Singh et al., 2017)
		Susceptible	Levofloxacin (levo)	Cell cytoplasm	Inhibits DNA gyrase and halts DNA replication	Noel (2009)
Gram-positive	*E. faecalis*	Susceptible			Inhibits DNA gyrase and topoisomerase IV to halt DNA replication	Kohanski et al. (2010)
		Resistant			Target modification	(Onodera et al., 2002; Korten et al., 1994)
	S. aureus	Susceptible	Bacitracin (bac)	Cell wall	Blocks cell wall formation by interfering with peptidoglycans	Umbreit (1953)
		Susceptible	Methicillin		Blocks penicillin-binding proteins (PBPs)	(Beale et al., 2010; Carretto et al., 2018)
		Resistant			Target modification	(Carretto et al., 2018; Peacock and Paterson, 2015)

sulfate (PMB) and sushi S3 peptide (sushi) were selected for their role in disrupting the outer cell membrane of Gram-negative bacteria (Poirel et al., 2017; Singh et al., 2017), while bacitracin (bac) and methicillin were shortlisted for their involvement in blocking the cell wall formation in Gram-positive bacteria (Umbreit, 1953; Beale et al., 2010). Both these sets of drugs also did not cross-react with the other Gram type and thus, provided good negative controls. In addition, levofloxacin (levo) was used as a common drug affecting both Gram-negative and Gram-positive bacteria as it is known to inhibit cell replication by breaking DNA double strands in both cell types (Noel, 2009). In this way, a comprehensive set of experiments was designed to test the efficacy of drugs that act on different parts of the bacteria (outer membrane *vs* cell wall *vs* cytoplasm). Including both susceptible and resistant strains further allowed us to explore the generality of our approach and establish its true limits of utility for a clinical application.

The first set of DEPIS studies were carried out on susceptible bacteria. Before proceeding with the actual experiments, we first performed standard culture tests using the plating method to determine the maximal bactericidal concentration (MBC_{100}) and half-maximal bactericidal concentration (MBC_{50}) of the drugs against any particular bacteria type. This was needed to benchmark our results with an existing gold standard, and also to narrow down the drug concentration range for

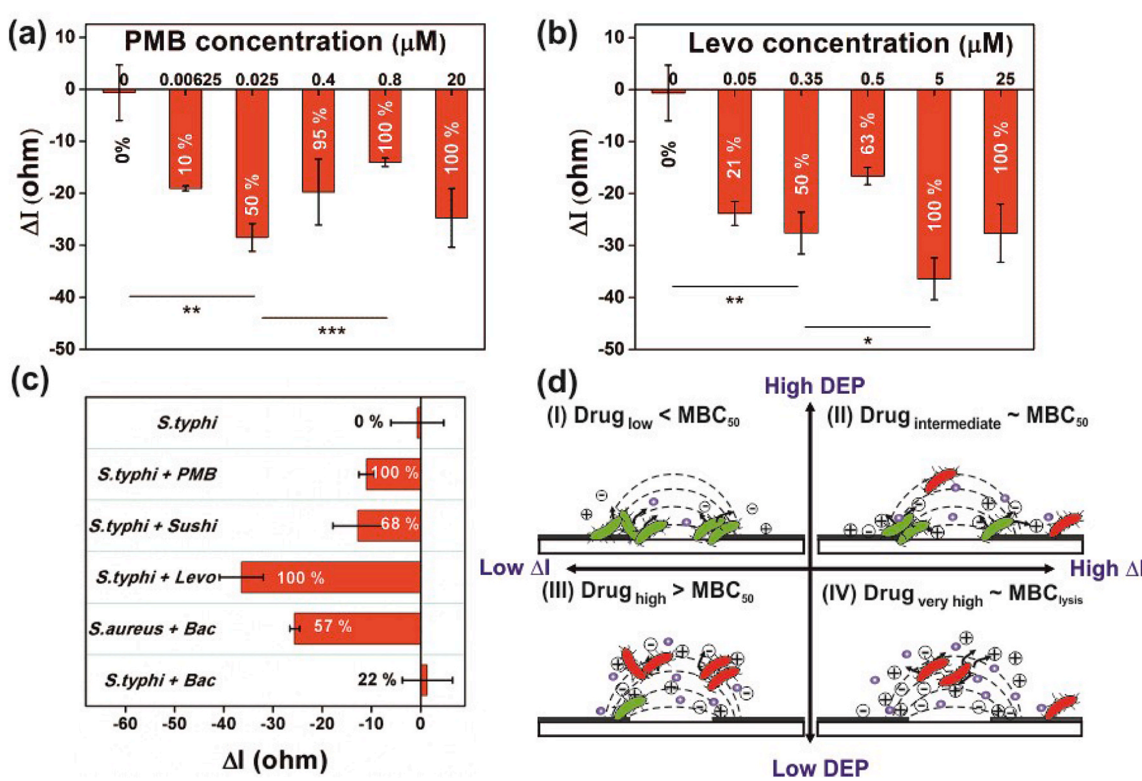

Fig. 3. One-hour dose response curves obtained using DEPIS platform for *S. typhi* bacteria (10^7 cfu/mL) in the presence of **(a)** PMB and **(b)** levo antibiotic. The values inside the bars indicate % cell death determined using the plating method. Also, 'ns' stands for not significant ($p > 0.05$) and *, **, *** imply $0.01 \leq p \leq 0.05$, $0.001 \leq p < 0.01$ and $p < 0.001$, respectively. **(c)** Impedance changes obtained for different cell-drug combinations at 5 μM drug concentration (also see Table 1). **(d)** Proposed phenomena for bacterial capture through DEP and ion concentration measurement as a function of antibiotic concentration in our chip through IS. Here, the ΔI response is small in *region I* (low ionic release at the electrodes with high cell capture) and *region III* (high ionic release in bulk at low cell capture), whereas, ΔI response is high in *region II* (moderate ionic release at the electrodes with moderate cell capture) and *region IV* (excessive ionic release in bulk without cell capture).

performing the DEPIS studies. The MBC values could not be directly borrowed from the literature as our experiments were performed over a much shorter time duration of drug exposure (60 min only) as opposed to the typical overnight incubation used in cell cultures. Once the concentration range for a given drug was identified, experiments were repeated in the DEPIS setup and the final results were plotted as differential impedance (ΔI) defined as the impedance measured at any time t = t' (where t' > 30 min) minus the impedance at time t = 0 (defined as the time when the drug was first mixed with the cells). The reason for plotting ΔI instead of the more frequently plotted parameters like I or $\frac{\Delta I}{I_{t=0}}$ (Yang, 2008; Valero et al., 2010) was that it was directly indicative of the ionic release in our system.

The 1 h DEPIS-MBC plots obtained for antibiotics PMB and levo against *S. typhi* bacteria are depicted in Fig. 3a and b, and their corresponding culture-MBC plots and $MBC_{50/100}$ values are reported in Figs. S6a and S6b & Table S2, respectively. Here, unlike the conventional sigmoidal shape of the culture-MBC plots, the DEPIS dose response curves were found to be highly non-monotonic in nature. The ΔI values for the buffer alone (zero drug concentration) were negligible. As the drug concentration was increased, the magnitude of the ΔI values also rose (actual sign was negative, i.e., the system became more conductive) up to MBC_{50} and then eventually decreased. Interestingly, the values did not become zero at MBC_{100}. At an even higher drug concentration where the cells underwent rapid lysis (MBC_{lysis}), again a relatively high signal was measured especially in case of PMB.

Furthermore, the results plotted in the frequency domain showed a maximum shift in signal near the 100 kHz range as predicted by our model in Fig. 2d (Figs. S7a and S7b). The drug-concentration related impedance events could, in fact, even be measured in the MHz frequency range where the bacterial cytoplasmic components start to contribute to the overall impedance but this was not pursued in our present study. Time-dependant MBC_{50} data revealed that changes in the solution resistance due to drug-mediated ionic release could be measured as early as 30 min into the assay (just after the end of the DEP step) and the signals continued to change even after 1 h indicating the sustained death of bacteria beyond this point (Figs. S7c, S7d, S8). These results indicated that our 1 h MBC plots comprised of kinetically-limited data, not true end-point values and DEP was thus indeed an important step since the ionic release took place continuously over the entire 60 min period. Further experiments performed with other drugs and bacteria showed that sushi generated a considerable shift in signal against *S. typhi.* whereas, bac (negative control) in comparison led to very little signal change due to its lack of affinity for Gram-negative bacteria (Fig. 3c). The interaction of bac with *S. aureus*, on the other hand, tested positive as anticipated from Table 1. Antibiotics on their own did not impact the buffer conductivity (Table S3). When analogous experiments were performed with UV–visible spectroscopy to map the changes in optical density to cell viability, we found the data to be completely unreliable (Fig. S9).

We believe that the non-monotonicity in our no liquid flow setup is caused by two competing phenomena: live bacterial cell capture by pDEP and the ionic release by cells near the electrodes. The extent of both these phenomena depend on the drug concentration used, which in turn affects the conductivity of the surrounding medium. The entire dose-response curve (Fig. 3a and b) can be understood by studying these two competing phenomena in detail as shown in Fig. 3d. At low antibiotic concentrations (<MBC_{50}), there is high pDEP capture as most cells are still alive but the ionic release is low (Fig. 3d–I). This results in very little overall impedance change in the system. At intermediate drug concentration (~MBC_{50}), there is relatively less pDEP capture as only 50% of the cells are alive, however, the cells that do get captured release overall more number of ions as compared to the previous case which leads to a bigger impedance change (Fig. 3d–II). At high drug concentrations (>MBC_{50}), very few cells get pDEP captured as a majority of them die in the bulk phase (Fig. 3d–III). So even though the ionic release is quite high, the ions are released far away from the detection zone and thus, there is a net decrease in the signal (Fig. 3d–III). Finally at extremely high drug concentrations (~MBC_{lysis}), most bacteria are lysed causing a deluge of cytoplasmic ions in the bulk phase. This makes the overall suspension highly conductive in spite of no actual bacteria trapped at the electrodes as there are none left alive to be captured (Fig. 3d–IV). This scenario again causes a sharp rise in the net signal as observed in our experiments. Therefore, the impedance cell viability curves follow a non-monotonic trend with two local minima, one at MBC_{50} and the other at MBC_{lysis}.

The reason why the magnitude of the ΔI response was greater in MBC_{lysis} than MBC_{100} in case of PMB, and the other way round in levo, is linked to the distinct ways in which these two drugs interact with the cells. PMB is a positively charged drug that kills Gram-negative bacteria by targeting the negatively charged lipopolysaccharide (LPS) molecules that are present there in abundance in the outer membrane (OM) of these cells. This leads to the formation of ion permeable pores that cause the slow release of K^+ and Mg^{2+} ions from the cytosol (Daugelavicius et al., 2000; Hirai et al., 1986). These K^+ and Mg^{2+} ions are present in the cytosol in much greater abundance than other ions. A back-of-the-envelope calculation of ionic release in our system showed excellent agreement between the theoretical and experimental values (see **SI**). At very high drug concentrations, the entire cytoplasmic membrane is destabilized leading to a flux of cytoplasmic ions, genomic content and proteins to be released into the medium (Krupovič et al., 2007; Schindler et al., 1975). This is what likely causes a high jump in signal at MBC_{lysis} in case of PMB.

Levo, on the other hand, is a second generation bactericidal fluoroquinolone drug that enters into the cell's cytoplasm by disrupting the OM LPS barrier (Delcour, 2010; Cramariuc et al., 2012) and cleaves the cell's DNA (Kohanski et al., 2010). This DNA fragmentation increases with the antibiotic incubation time and concentration, and diffuses from the nucleoid into the medium (Tamayo et al., 2009). Although there are no drug concentration-dependent ionic release studies cited directly for levo, based on the literature and our own data, we deduced that the cell-drug interaction leads to an immediate destabilization of the membrane along with the release of bacterial cytoplasmic components (ions, negatively charged DNA fragments, DNA binding proteins *etc.*) that increase the medium conductivity. This is the reason why the ionic release was quite high for levo even at MBC_{100} and when compared to other drugs taken at the same concentration (see Fig. 3c). Since we report all our data in terms of ΔI, for a drug that causes immediate ionic release, a higher starting value of impedance at t = 0 would lead to smaller ΔI response at very high drug concentrations (since substantial ionic release would take place around at t = 0 itself when the antibiotic is first added, making ΔI effectively smaller). This is possibly the reason why the signal slightly dropped at MBC_{lysis} in case of levo. Finally, we would also like to point out that while the magnitudes of impedance signal were seen to be comparable in levo and PMB, the potency (defined as the amount of drug required to produce an effect of a given intensity) of PMB was found to be much higher than levo as expected from the reported MIC values in the literature (see Table S2).

Finally, to explicitly rule out the effect of electroporation, which can become significant at high voltages such as the one used for DEP (20 V_{pp}) (Pavlin et al., 2005), we compared the impedance data for cell suspensions at low frequencies with and without DEP (Castellví et al., 2016). Electroporation increases the cell membrane permeability at low frequency causing cytoplasmic ion release through the membrane and, thus leading to a decrease in the overall impedance signal. This effect, however, can not be measured at higher frequencies since at high frequencies, current flows freely across the membrane at all conditions (or in other words, the membrane is short-circuited). We found out that there was indeed a discernible drop in impedance between 1 and 10 kHz (Fig. S10), hinting that electroporation might be a contributing factor in the first 30 min of ionic release (on top of ionic release by antibiotics) when DEP was applied. However, Fig. 4c showed no significant

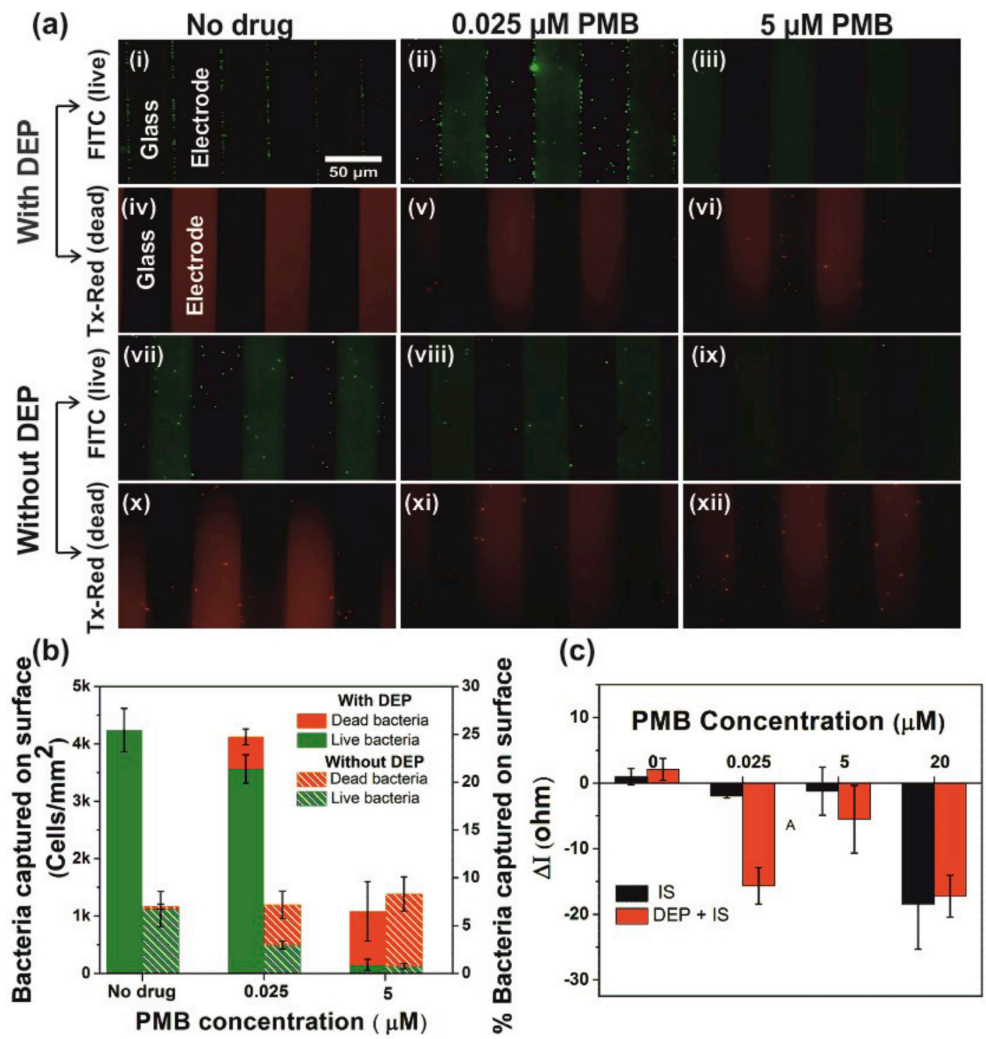

Fig. 4. Microscopy analyses and role of DEP. **(a)** Fluorescence images depicting live and dead *S. typhi* bacteria collected on the bottom electrode surface with or without DEP and in the presence of different PMB drug concentrations (No drug, MBC_{50} = 0.025 μM and MBC_{100} = 5 μM). The live bacteria appear green under the FITC filter and the dead bacteria appear red using the Texas-red filter. The top two panels are complementary to each other as are the bottom two panels except the first two images of the first column (with DEP no drug case). In the FITC panels, the black regions represent the electrodes and green regions indicate glass, whereas, in the Tx-red rows, red stands for the electrode and black is glass. **(b)** Quantitative analyses of (a) after averaging over six such images. Here, FOV per image is equal to 37000 μm^2. **(c)** Comparative impedance data obtained with and without DEP at various PMB concentrations highlighting the essential role of DEP in making the process more sensitive. All data are reported at 100 kHz after 30 min of drug exposure. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

difference in ionic release with/without DEP in the absence of drug, which led us to conclude that electroporation effects were negligible in our system, and any drop in impedance values at lower frequencies was likely due to increase in local cell concentration near the electrodes (Yang et al., 2008). An indirect effect of electroporation via membrane destabilization, aiding in higher antibiotic penetration/action may be present, however, this could not be directly verified by our experiments.

3.1.2. Microscopy analyses and the role of DEP

Our MBC results were further cross-verified with detailed microscopy analysis. For this, the bacteria were first stained with Syto 9 and PI dyes using a LIVE/DEAD cell viability kit. The cells were then captured via pDEP in the presence of different antibiotic concentrations and the number of live-dead bacteria captured on the surface was quantified using time-lapse fluorescence microscopy. The images captured over 30 min revealed that as the antibiotic concentration in the system was increased, the number of live bacteria captured at the electrode surface decreased (Fig. 4a). This happened mainly due to two reasons: (i) the number of bacteria that remained alive at higher drug concentrations was less, and (ii) active cell death led to an increase in the medium conductivity which in turn led to lower capture of live cells. When we calculated the medium conductivity after the cellular ionic release, its value came out to be only 0.044 S/m (see **SI**) which was comparable to the conductivity of the buffer alone (0.04 S/m). Thus, we could safely say that the only reason for less bacterial capture at the electrodes at high drug concentrations was less number of live bacteria remaining in the suspension. At the EF condition used, the polarizability of the dead bacteria were comparable to that of the medium and so the dead cells settled mainly due to gravity, not nDEP (CM factor of - 0.24 implied small nDEP force). This was evident from Fig. 4a–vi which showed a relatively large number of dead bacteria uniformly scattered on the bottom surface, as opposed to the live bacteria that normally get collected at the electrode edges. Also, the number of dead bacteria collected with and without DEP was quite similar indicating that there was little role of nDEP. What was also interesting was the fact that while the bacteria exposed to MBC_{50} (Fig. 4-ii) still appeared green, or in other words alive, they were no longer restricted to the electrode edges and were instead more spread out between the electrode fingers. This suggested that the cells were not completely dead, but rather in a slow dying phase, and were thus able to release ions near the electrodes as claimed in Fig. 3d–II. At the highest drug concentration of 20 μM (MBC_{lysis}), even fewer dead bacteria were found at the electrode surface as proposed in Fig. 3d–IV (Fig. S11) suggesting that most of the cells got lysed completely in the bulk phase.

To confirm the relevance of DEP in our approach, we also repeated the above experiments in the absence of DEP. The qualitative results obtained are shown in the bottom two panels in Fig. 4a and the quantitative comparisons are mentioned in Fig. 4b. The overall trends were found to be quite similar with and without DEP with the exceptions that the number of cells collected without DEP was 4x less than that using DEP, and the bacteria were more randomly distributed throughout the bottom surface. Both these factors contributed greatly to the significant lowering of impedance signal in the absence of DEP (Fig. 4c), highlighting the crucial role of the capture step in our method. The maximum impact of DEP was seen at MBC_{50} and its effect was found to be the least at MBC_{lysis} where there were very few intact bacterial cells left to be captured.

3.1.3. Rapid differentiation of resistant vs susceptible bacteria

The purpose of any AST device is to guide clinical therapy. This includes rapid differentiation of susceptible and resistant bacteria for administrating the right antibiotic. We tested the crucial functionality of resistance profiling in our platform by using two reference bacterial strains - methicillin-resistant *S. aureus* (MRSA) and levo-resistant *E. faecalis* (EFR) and their corresponding antibiotics methicillin and levo, respectively. The reason why these two drugs were chosen was because the target-modification due to resistance against these antibiotics occurs in very different regions of the cells. The resistance to levo in EFR happens due to the mutation of *gyrA and parC* genes that code for DNA gyrase and topoisomerase IV, respectively (Onodera et al., 2002; Korten et al., 1994). This target modification inhibits the binding of levo to these enzymes *inside* the cytoplasm. Methicillin, on the other hand, acts on the bacterial *surface* by blocking and disrupting penicillin-binding proteins (PBPs) that synthesize peptidoglycans, the essential constituents of the bacterial cell wall. Modification of the *mecA* gene in PBPs (also known as foreign PBPs) lowers their compatibility for methicillin binding and manifests as resistance in MRSA, no longer able to stop their growth (Carretto et al., 2018; Peacock and Paterson, 2015).

Having both target-modified resistant bacteria MRSA and EFR, impedance experiments were performed by exposing these cells to 0.2 μM methicillin and 0.5 μM levo, respectively. Control experiments were conducted on susceptible *S. aureus* (MSSA) and susceptible *E. faecalis* (EFS). The results plotted in Fig. 5a showed that the net impedance change was similar in EFS and EFR even though the extent of cell death was significantly different in these two strains. On the other hand, MSSA and MRSA showed a sharp difference in impedance response in proportion to the extent of cell death (Fig. 5b). In our opinion, the reason for the smaller ionic release in MRSA compared to MSSA is linked to the reduced binding affinity of methicillin to foreign PBP enzymes present on the cell surface. Levo, on the other hand, is a cytoplasmic antibiotic that has to penetrate the cell wall in order to act. The entry of this molecule into the cytoplasm is what likely causes the loss of intracellular ions irrespective of the internal resistance status of the bacteria. From these results we conclude that our platform can only effectively differentiate resistant bacteria from susceptible ones for antibiotics that target the cell wall. Extended studies are now underway with more clinical strains to fully validate our claims.

4. Conclusions

In summary, we have demonstrated a simple and affordable label-free biosensor that gives MBC_{50} values of bactericidal antibiotics in just 30–60 min. The experiments were performed with reusable electrodes that require no prior functionalization or fluid flow. Tests performed with reference strains of *S. typhi, E. faecalis* and *S. aureus* and five different antibiotics (PMB, sushi, levo, bac and methicillin) showed that the sensor works well for Gram-negative and Gram-Positive bacteria alike and is highly selective. Moreover, it does not depend on the action mechanism or cellular target of the drug type. Experiments performed with MRSA and EFR revealed that resistant and susceptible bacterial strains can be easily differentiated in only an hour as long as the resistance mechanism is linked to modified binding targets at the cell wall. This potentially includes other broad-spectrum β-lactam antibiotics that are regarded as the last-resort medicine. Among low conductivity buffers, 15 mM HEPES at pH 7.4 (0.04 S/m) was found to be most suitable for our studies as it gave high pDEP cell capture, high impedance signal, good buffering capacity, and the least amount of stress to the cells in the absence of drugs. The pDEP force was applied for 30 min at 20 Vpp and 5 MHz to obtain approx. 4250 cells/mm^2 which was sufficient to elicit a measurable impedance response even at nanomolar drug concentrations. The most appropriate frequency for impedance studies was found to be 100 kHz as here, the solution resistance dominated the overall impedance behaviour of the system and therefore, yielded maximum signal change due to ionic release by the dying bacteria. Optical microscopy data supported our impedance observations and highlighted the essential role of DEP in making the process ultra-sensitive. The overall conclusion from our study is that ionic release from dying cells is a good proxy indicator of their viability and one of the fastest ways to generate bactericidal antibiotic dose response for drug susceptibility profiling in real-time. One limitation of our current approach is that it does not cater to bacteriostatic antibiotics due to the no growth medium used. However, this limitation can be overcome through the use of low conductivity growth buffers that allow measuring impedance changes mainly due to increase in biomass (over 2 to 3 cell divisions) and partially due to antibiotic-induced ionic release depending on the mechanism of drug uptake. This work in currently under progress in our lab.

CRediT authorship contribution statement

Pragya Swami: Methodology, Investigation, Formal analysis, Writing – original draft, Visualization. **Ayush Sharma:** Investigation, Formal analysis, Writing – original draft, Visualization. **Satyam Anand:** Software, Formal analysis, Writing – original draft. **Shalini Gupta:** Conceptualization, Supervision, Resources, Writing – review & editing, Visualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial

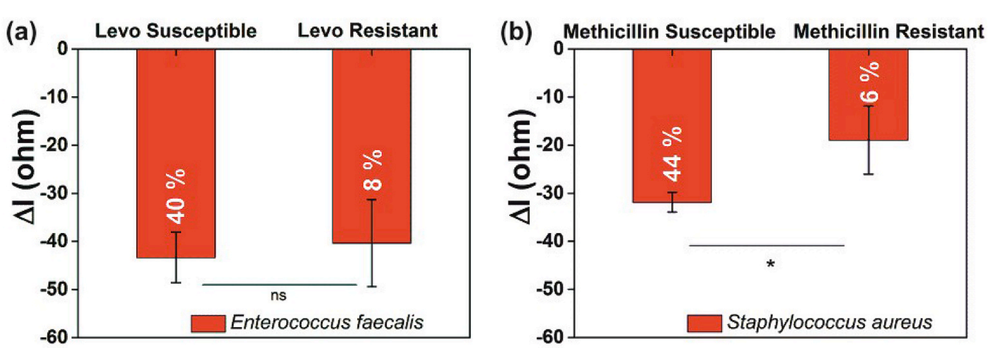

Fig. 5. Antibiotic susceptibility profiling. (a) Impedance data for levo-susceptible and levo-resistant *E. faecalis* after 1 h of drug exposure (0.5 μM levo). Here, the lack of signal change highlights that our method cannot differentiate resistance developed against drugs that act inside the cytoplasm. **(b)** Impedance data for methicillin-susceptible and methicillin-resistant *S. aureus* after 1 h of drug exposure (0.2 μM methicillin). The significant difference in signal indicates the efficacy of our approach in differentiating resistance developed against drugs that act on the cell wall. The results are plotted at 100 kHz and the values reported inside each bar graph indicate the % bacterial cell death at that condition as determined by the plating method. 'ns' stands for not significant ($p > 0.05$) and '*' implies $0.01 \leq p \leq 0.05$.

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by the funds received from DST Nanomission (SR/NM/NT-1049/2016) and IMPRINT (IMP/2018/000236/HT). We are grateful to Vikas Manchanda from Maulana Azad Medical College, New Delhi, Venkataraman Sritharan from Global hospitals, Hyderabad and Atul Narang from Dept. of Biochemical and Bioengineering, IIT Delhi for providing the bacterial strains. We also thank Nathan Swami from Univ. of Virginia, USA for helpful discussions. PS acknowledges the fellowship received from IIT Delhi.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bios.2021.113190.

References

Barbero, G., Alexe-Ionescu, A.L., Lelidis, I., 2005. Significance of small voltage in impedance spectroscopy measurements on electrolytic cells. J. Appl. Phys. 98, 113703. https://doi.org/10.1063/1.2137444.

Beale, J.M., Block, J., Hill, R., 2010. Organic Medicinal and Pharmaceutical Chemistry, twelfth ed. Lippincott Williams & Wilkins.

Carretto, E., Visiello, R., Nardini, P., 2018. Methicillin resistance in Staphylococcus aureus. Pet-to-Man travel. Staphylococci A World Prog 85, 225–235. https://doi.org/10.1016/B978-0-12-813547-1.00017-0.

Castellví, Q., Mercadal, B., Ivorra, A., 2016. Handbook of electroporation. Handb. Electroporation 1–20. https://doi.org/10.1007/978-3-319-26779-1.

Chantell, C., 2015. Multiplexed automated digital microscopy for rapid identification and antimicrobial susceptibility testing of bacteria and yeast directly from clinical samples. Clin. Microbiol. Newsl. 37, 161–167. https://doi.org/10.1016/j.clinmicnews.2015.10.001.

Christiansen, J., 2018. Global infections by the numbers. Sci. Am. 318, 48–49. https://doi.org/10.1038/scientificamerican0518-44.

Choi, J., Jung, Y.G., Kim, J., Kim, S., Jung, Y., Na, H., Kwon, S., 2013. Rapid antibiotic susceptibility testing by tracking single cell growth in a microfluidic agarose channel system. Lab Chip 13, 280–287. https://doi.org/10.1039/c2lc41055a.

Cramariuc, O., Rog, T., Javanainen, M., Monticelli, L., Polishchuk, A.V., Vattulainen, I., 2012. Mechanism for translocation of fluoroquinolones across lipid membranes. Biochim. Biophys. Acta Biomembr. 1818, 2563–2571. https://doi.org/10.1016/j.bbamem.2012.05.027.

Couniot, N., Afzalian, A., Van Overstraeten-Schlögel, N., Francis, L.A., Flandre, D., 2015. Capacitive biosensing of bacterial cells: analytical model and numerical simulations. Sensor. Actuator. B Chem. 211, 428–438. https://doi.org/10.1016/j.snb.2015.01.108.

Couniot, N., Bol, D., Poncelet, O., Francis, L.A., Flandre, D., 2014. A capacitance-to-frequency converter with on-chip passivated microelectrodes for bacteria detection in saline buffers up to 575 MHz. IEEE Trans. Circuits Syst. II Express Briefs 62, 159–163. https://doi.org/10.1109/TCSII.2014.2369111.

Dastider, S.G., Abdullah, A., Jasim, I., Yuksek, N.S., Dweik, M., Almasri, M., 2018. Low concentration E. coli O157:H7 bacteria sensing using microfluidic MEMS biosensor. Rev. Sci. Instrum. 89 https://doi.org/10.1063/1.5043424.

Dastider, S.G., Barizuddin, S., Dweik, M., Almasri, M., 2013. A micromachined impedance biosensor for accurate and rapid detection of E. Coli O157:H7. RSC Adv. 3, 26297–26306. https://doi.org/10.1039/c3ra44724c.

Dastider, S.G., Barizuddin, S., Yuksek, N.S., Dweik, M., Almasri, M.F., 2015. Efficient and rapid detection of salmonella using microfluidic impedance based sensing. J. Sensors 2015. https://doi.org/10.1155/2015/293461.

Daugelavicius, R., Bakiene, E., Bamford, D.H., 2000. Stages of polymyxin B interaction with the Escherichia coli cell envelope. Antimicrob. Agents Chemother. 44, 2969–2978. https://doi.org/10.1128/AAC.44.11.2969-2978.2000.

Delcour, A.H., 2010. Outer membrane permeability and antibiotic resistance. Biochim. Biophys. Acta Protein Proteonomics 1794, 808–816. https://doi.org/10.1016/j.bbapap.2008.11.005.

Dietvorst, J., Vilaplana, L., Uria, N., Marco, M.P., Muñoz-Berbel, X., 2020. Current and near-future technologies for antibiotic susceptibility testing and resistant bacteria detection. Trac. Trends Anal. Chem. 127, 1–13. https://doi.org/10.1016/j.trac.2020.115891.

Felmingham, D., Brown, D.F.J., 2001. Instrumentation in antimicrobial susceptibility testing. Antimicrob. Chemother. 48, 81–85. https://doi.org/10.1093/jac/48.suppl_1.81.

Fluit, A.C., Visser, M.R., Schmitz, F.J., 2001. Molecular detection of antimicrobial resistance. Clin. Microbiol. Rev. 14, 836–871. https://doi.org/10.1128/CMR.14.4.836-871.2001.

Fredborg, M., Andersen, K.R., Droce, A., Olesen, T., Jensen, B.B., Rosenvinge, F.S., Sondergaard, T.E., 2013. Real-time optical antimicrobial susceptibility testing. J. Clin. Microbiol. 51, 2047–2053. https://doi.org/10.1128/JCM.00440-13.

Furst, A.L., Francis, M.B., 2019. Impedance-based detection of bacteria. Chem. Rev. 119, 700–726. https://doi.org/10.1021/acs.chemrev.8b00381.

Gallo-Villanueva, R.C., Sano, M.B., Lapizco-Encinas, B.H., Davalos, R.V., 2014. Joule heating effects on particle immobilization in insulator-based dielectrophoretic devices. Electrophoresis 35, 352–361. https://doi.org/10.1002/elps.201300171.

Gnaim, R., Golberg, A., Sheviryov, J., Rubinsky, B., González, C.A., 2020. Detection and differentiation of bacteria by electrical bioimpedance spectroscopy. Biotechniques 69, 27–36. https://doi.org/10.2144/BTN-2019-0080.

Godin, M., Bryan, A.K., Burg, T.P., Babcock, K., Manalis, S.R., 2007. Measuring the mass, density, and size of particles and cells using a suspended microchannel resonator. Appl. Phys. Lett. 91, 17–19. https://doi.org/10.1063/1.2789694.

Goel, M., Verma, A., Gupta, S., 2018. Electric-field driven assembly of live bacterial cell microarrays for rapid phenotypic assessment and cell viability testing. Biosens. Bioelectron. 111, 159–165. https://doi.org/10.1016/j.bios.2018.04.005.

Gomez-Sjoberg, R., Morisette, D.T., Bashir, R., 2005. Impedance microbiology-on-a-chip : microfluidic bioprocessor for rapid detection of bacterial metabolism. J. Microelectromech Syst. 14, 829–838. https://doi.org/10.1109/JMEMS.2005.845444.

Hirai, K., Aoyama, H., Irikura, T., Iyobe, S., Mitsuhashi, S., 1986. Differences in susceptibility to quinolones of outer membrane mutants of Salmonella typhimurium and Escherichia coli. Antimicrob. Agents Chemother. 29, 535–538. https://doi.org/10.1128/AAC.29.3.535.

Hou, T.Y., Chiang-Ni, C., Teng, S.H., 2019. Current status of MALDI-TOF mass spectrometry in clinical microbiology. J. Food Drug Anal. 27, 404–414. https://doi.org/10.1016/j.jfda.2019.01.001.

Johnson, M.B., Criss, A.K., 2013. Fluorescence microscopy methods for determining the viability of bacteria in association with mammalian cells. JoVE 79, 1–9. https://doi.org/10.3791/50729.

Jorgensen, J.H., Ferraro, M.J., 2009. Antimicrobial susceptibility Testing : a review of general principles and contemporary practices. Clin. Infect. Dis. 7750, 1749–1755. https://doi.org/10.1086/647952.

Khandelwal, A.V., Singh, A., Pal, N., Kumar, R., Goel, G., Gupta, S., 2019. AC conductivity measurements of ultradilute colloidal suspensions in HEPES buffer. Langmuir 35, 14725–14733. https://doi.org/10.1021/acs.langmuir.9b01464.

Kohanski, M.A., Dwyer, D.J., Collins, J.J., 2010. How antibiotics kill bacteria: from targets to networks. Nat. Rev. Microbiol. 8, 423–435. https://doi.org/10.1038/nrmicro2333.

Korten, V., Huang, W.M., Murray, B.E., 1994. Analysis by PCR and direct DNA sequencing of gyrA mutations associated with fluoroquinolone resistance in Enterococcus faecalis. Antimicrob. Agents Chemother. 38, 2091–2094. https://doi.org/10.1128/AAC.38.9.2091.

Krupovič, M., Daugelavičius, R., Bamford, D.H., 2007. Polymyxin B induces lysis of marine pseudoalteromonads. Antimicrob. Agents Chemother. 51, 3908–3914. https://doi.org/10.1128/AAC.00449-07.

Lei, K.F., 2014. Review on impedance detection of cellular responses in micro/nano environment. Micromachines 5, 1–12. https://doi.org/10.3390/mi5010001.

Leva-Bueno, J., Peyman, S.A., Millner, P.A., 2020. A review on impedimetric immunosensors for pathogen and biomarker detection. Med. Microbiol. Immunol. 209, 343–362. https://doi.org/10.1007/s00430-020-00668-0.

Li, B., Qiu, Y., Glidle, A., McIlvenna, D., Luo, Q., Cooper, J., Shi, H.C., Yin, H., 2014. Gradient microfluidics enables rapid bacterial growth inhibition testing. Anal. Chem. 86, 3131–3137. https://doi.org/10.1021/ac5001306.

Li, H., Multari, C., Palego, C., Ma, X., Du, X., Ning, Y., Buceta, J., Hwang, J.C., Cheng, X., 2018. Differentiation of live and heat-killed E. coli by microwave impedance spectroscopy. Sensor. Actuator. B Chem. 255, 1614–1622. https://doi.org/10.1016/j.snb.2017.08.179.

Lillehoj, P.B., Kaplan, C.W., He, J., Shi, W., Ho, C.M., 2014. Rapid, electrical impedance detection of bacterial pathogens using immobilized antimicrobial peptides. J. Lab. Autom. 19, 42–49. https://doi.org/10.1177/2211068213495207.

Liu, J., Jasim, I., Abdullah, A., Shen, Z., Zhao, L., El-Dweik, M., Zhang, S., Almasri, M., 2018. An integrated impedance biosensor platform for detection of pathogens in poultry products. Sci. Rep. 8, 1–10. https://doi.org/10.1038/s41598-018-33972-0.

Liu, Y.Q., Zhang, Y.Z., Gao, P.J., 2004. Novel concentration-killing curve method for estimation of bactericidal potency of antibiotics in an in vitro dynamic model. Antimicrob. Agents Chemother. 48, 3884–3891. https://doi.org/10.1128/AAC.48.10.3884. Liu et al., 2004.

Llor, C., Bjerrum, L., 2014. Antimicrobial resistance: risk associated with antibiotic overuse and initiatives to reduce the problem. Ther. Adv. Drug Saf. 5, 229–241. https://doi.org/10.1177/2042098614554919.

Longo, G., Rio, L.M., Bizzini, A., Trampuz, A., Notz, J., Dietler, G., Kasas, S., 2013. Rapid detection of bacterial resistance to antibiotics using AFM cantilevers as nanomechanical sensors. Nat. Nanotechnol. 8, 522. https://doi.org/10.1038/nnano.2013.120.

Mallén-Alberdi, M., Vigués, N., Mas, J., Fernández-Sánchez, C., Baldi, A., 2016. Impedance spectral fingerprint of E. coli cells on interdigitated electrodes: a new approach for label free and selective detection. Sens. Bio-Sensing Res. 7, 100–106. https://doi.org/10.1016/j.sbsr.2016.02.001.

Mannoor, M.S., Zhang, S., Link, A.J., McAlpine, M.C., 2010. Electrical detection of pathogenic bacteria via immobilized antimicrobial peptides. Proc. Natl. Acad. Sci. U. S.A. 107, 19207–19212. https://doi.org/10.1073/pnas.1008768107.

McDonald, J.C., Duffy, D.C., Anderson, J.R., Chiu, D.T., Wu, H., Schueller, O.J.A., Whitesides, G.M., 2000. Fabrication of microfluidic systems in poly (dimethylsiloxane). Electrophoresis 21, 27–40. https://doi.org/10.1002/(sici)1522-2683(20000101)21:1<27::aid-elps27>3.3.co;2-3.

Mohan, R., Mach, K.E., Bercovici, M., Pan, Y., Dhulipala, L., Wong, P.K., Liao, J.C., 2011. Clinical validation of integrated nucleic acid and protein detection on an

electrochemical biosensor array for urinary tract infection diagnosis. PloS One 6. https://doi.org/10.1371/journal.pone.0026846.
Moral-zamora, B., Punter-villagrassa, J., Oliva-bra, A.M., Alvarez, J.M., Colomer-farrarons, J., Samitier, J., Homs-corbera, A., 2015. Combined dielectrophoretic and impedance system for on-chip controlled bacteria concentration : application to Escherichia coli. Electrophoresis 36, 1130–1141. https://doi.org/10.1002/elps.201400446.
Noel, G.J., 2009. A review of levofloxacin for the treatment of bacterial infections. Clin. Med. Therapeut. 1 https://doi.org/10.4137/cmt.s28. CMT.S28.
O'Neill, J., 2014. Antimicrobial resistance: tackling a crisis for the health and wealth of nations. Rev. Antimicrob. Resist. 20, 1–16.
Onodera, Y., Okuda, J., Tanaka, M., Sato, K., 2002. Inhibitory activities of quinolones against DNA gyrase and topoisomerase IV of Enterococcus faecalis. Antimicrob. Agents Chemother. 46, 1800–1804. https://doi.org/10.1128/AAC.46.6.1800-1804.2002.
Pal, N., Sharma, S., Gupta, S., 2016. Sensitive and rapid detection of pathogenic bacteria in small volumes using impedance spectroscopy technique. Biosens. Bioelectron. 77, 270–276. https://doi.org/10.1016/j.bios.2015.09.037.
Pavlin, M., Kandušer, M., Reberšek, M., Pucihar, G., Hart, F.X., Magjarević, R., Miklavčič, D., 2005. Effect of cell electroporation on the conductivity of a cell suspension. Biophys. J. 88, 4378–4390. https://doi.org/10.1529/biophysj.104.048975.
Peacock, S.J., Paterson, G.K., 2015. Mechanisms of methicillin resistance in Staphylococcus aureus. Annu. Rev. Biochem. 84, 577–601. https://doi.org/10.1146/annurev-biochem-060614-034516.
Pethig, R., 2010. Review article — dielectrophoresis : status of the theory, technology, and applications. Biomicrofluidics 4, 022811. https://doi.org/10.1063/1.3456626.
Poirel, L., Jayol, A., Nordmann, P., 2017. Polymyxins: antibacterial activity, susceptibility testing, and resistance mechanisms encoded by plasmids or chromosomes. Clin. Microbiol. Rev. 30, 557–596. https://doi.org/10.1128/CMR.00064-16.
Rani, G., Patri, I., 2019. Importance of being cross-linked for the bacterial cell wall. Phys. Rev. E 100, 1–13. https://doi.org/10.1103/PhysRevE.100.062408.
Rohani, A., Moore, J.H., Su, Y.H., Stagnaro, V., Warren, C., Swami, N.S., 2018. Single-cell electro-phenotyping for rapid assessment of Clostridium difficile heterogeneity under vancomycin treatment at sub-MIC (minimum inhibitory concentration) levels. Sensor. Actuator. B Chem. 276, 472–480. https://doi.org/10.1016/j.snb.2018.08.137.
Sabounchi, P., Morales, A.M., Ponce, P., Lee, L.P., Simmons, B.A., Davalos, R.V., 2008. Sample concentration and impedance detection on a microfluidic polymer chip. Biomed. Microdevices 10, 661–670. https://doi.org/10.1007/s10544-008-9177-4.
Safavieh, M., Pandya, H.J., Venkataraman, M., Thirumalaraju, P., Kanakasabapathy, M.K., Singh, A., Prabhakar, D., Chug, M.K., 2017. Rapid real-time antimicrobial susceptibility testing with electrical sensing on plastic microchips with printed electrodes. ACS Appl. Mater. Interfaces 9, 12832–12840. https://doi.org/10.1021/acsami.6b16571.
Schindler, P.R.G., Teuber, M., 1975. Action of polymyxin B on bacterial membranes: morphological changes in the cytoplasm and in the outer membrane of Salmonella typhimurium and Escherichia coli B. Antimicrob. Agents Chemother. 8, 95–104. https://doi.org/10.1128/AAC.8.1.95.
Sinn, I., Albertson, T., Kinnunen, P., Breslauer, D.N., McNaughton, B.H., Burns, M.A., Kopelman, R., 2012. Asynchronous magnetic bead rotation (AMBR) microviscometer for rapid, sensitive and label-free studies of bacterial growth and drug sensitivity. Anal. Chem. 84, 5250–5256. https://doi.org/10.1021/ac300128p.Asynchronous.
Singh, R., Patil, S., Singh, N., Gupta, S., 2017. Dual functionality nanobioconjugates targeting intracellular bacteria in cancer cells with enhanced antimicrobial activity. Sci. Rep. 7, 1–10. https://doi.org/10.1038/s41598-017-06014-4.
Spencer, D.C., Paton, T.F., Mulroney, K.T., Inglis, T.J.J., Sutton, J.M., Morgan, H., 2020. A fast impedance-based antimicrobial susceptibility test. Nat. Commun. 11 https://doi.org/10.1038/s41467-020-18902-x.
Suehiro, J., Ohtsubo, A., Hatano, T., Hara, M., 2006. Selective detection of bacteria by a dielectrophoretic impedance measurement method using an antibody-immobilized electrode chip. Sensor. Actuator. B Chem. 119, 319–326. https://doi.org/10.1016/j.snb.2005.12.027.
Sun, T., Green, N.G., Morgan, H., 2008. Analytical and numerical modeling methods for impedance analysis of single cells on-chip. Nano 55–63. https://doi.org/10.1142/s1793292008000800, 03.
Suo, Z., Yang, X., Avci, R., Kellerman, L., Pascual, D.W., Fries, M., Steele, A., 2007. HEPES-stabilized encapsulation of Salmonella typhimurium. Langmuir 23, 1365–1374. https://doi.org/10.1021/la0621721.
Superbug Solutions Platform, 2019. http://superbugsolutions.webflow.io/.
Syal, K., Mo, M., Yu, H., Iriya, R., Jing, W., Guodong, S., Wang, S., Grys, T.E., Haydel, S.E., Tao, N., 2017. Current and emerging techniques for antibiotic susceptibility tests. Theranostics 7, 1795–1805. https://doi.org/10.7150/thno.19217.
Tamayo, M., Santiso, R., Gosalvez, J., Bou, G., Fernández, J.L., 2009. Rapid assessment of the effect of ciprofloxacin on chromosomal DNA from Escherichia coli using an in situ DNA fragmentation assay. BMC Microbiol. 9, 1–11. https://doi.org/10.1186/1471-2180-9-69.
Ten, T., Killers, B., Ten, T., Common, M., Diseases, N., 2017. World health report Press release 1995–1996.
Umbreit, W.W., 1953. Mechanisms of antibacterial action. Pharmacol. Rev. 5, 275–284. https://doi.org/10.1146/annurev.mi.08.100154.001123.
Valero, A., Braschler, T., Renaud, P., 2010. A unified approach to dielectric single cell analysis: impedance and dielectrophoretic force spectroscopy. Lab Chip 10, 2216–2225. https://doi.org/10.1039/c003982a.
Ventola, C.L., 2015. The antibiotic resistance crisis: part 1: causes and threats. P T A peer-reviewed. J. Formul. Manag. 40, 277–283.
Webb, J.S., Barratt, S.R., Sabev, H., Nixon, M., Eastwood, I.M., Greenhalgh, M., Handley, P.S., Robson, G.D., 2001. Green fluorescent protein as a novel indicator of antimicrobial susceptibility in aureobasidium pullulans. Appl. Environ. Microbiol. 67, 5614–5620. https://doi.org/10.1128/AEM.67.12.5614-5620.2001.
Xu, Y., Xie, X., Duan, Y., Wang, L., Cheng, Z., Cheng, J., 2016. A review of impedance measurements of whole cells. Biosens. Bioelectron. 77, 824–836. https://doi.org/10.1016/j.bios.2015.10.027.
Yang, L., 2008. Electrical impedance spectroscopy for detection of bacterial cells in suspensions using interdigitated microelectrodes. Talanta 74, 1621–1629. https://doi.org/10.1016/j.talanta.2007.10.018.
Yang, L., Bashir, R., 2008. Electrical/electrochemical impedance for rapid detection of foodborne pathogenic bacteria. Biotechnol. Adv. 26, 135–150. https://doi.org/10.1016/j.biotechadv.2007.10.003.